2023 AMC 10A Problem 14 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 14. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 14

Problem:

A number is chosen at random from among the first 100 positive integers, and a positive integer divisor of that number is then chosen at random. What is the probability that the chosen divisor is divisible by 11?

Answer Choices:

A. $\frac{4}{100}$

B. $\frac{9}{200}$

C. $\frac{1}{20}$

D. $\frac{11}{200}$

E. $\frac{3}{50}$

Solution:

There are 9 multiples of 11 among the first 100 positive integers, and the probability of choosing any particular one of them is $\frac{1}{100}$. For each of these multiples of 11 , the probability that\
a randomly chosen divisor is a multiple of 11 is $\frac{1}{2}$ because each divisor that is a multiple of 11 can be paired with a divisor that is not a multiple of 11 . Therefore the requested probability is

$$\frac{9}{100} \cdot \frac{1}{2} = \text{(B)}\boxed{\frac{9}{200}}$$